Interest Rates and Real Private Fixed Investment Growth in the United States

ECON 3916 Midterm Project

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Question and Data

How is the annual average federal funds rate associated with annual real private fixed investment growth in the United States from 1968–2024, controlling for real GDP growth and inflation?

This question matters to businesses planning investment and policymakers assessing economic conditions. Interest rates influence financing costs, but investment also responds to broader economic conditions. A comparison that ignores these conditions may give an incomplete picture of the relationship.

Using 57 annual observations, I find that the estimated interest-rate coefficient changes across model specifications and is not statistically significant at the 5% level in any version. The full model produces a negative estimate, but its uncertainty prevents a strong conclusion about a negative conditional association. The analysis does not identify a causal effect.

Data Sources and Measurement

The data were downloaded from FRED, maintained by the Federal Reserve Bank of St. Louis. The outcome comes from the annual Real Private Fixed Investment series, FPICA. I calculate investment growth as the annual percentage change in the inflation-adjusted investment level. The outcome therefore measures growth in real private fixed investment rather than nominal spending growth.

The main explanatory variable is the federal funds rate, FEDFUNDS. Because this series is monthly, I average its observations within each year. The controls are annual real GDP growth, A191RL1A225NBEA, and annual inflation, FPCPITOTLZGUSA.

These measurement choices affect interpretation. Annual averages summarize interest-rate conditions over a year but do not capture the timing of individual rate changes. Investment growth may also respond to earlier financing conditions, which the contemporaneous models do not examine.

Sample Selection and Missingness

The downloaded investment file begins in 1967. Calculating growth requires a preceding-year investment value, so growth is first available in 1968. After combining the four series, the final sample contains 57 annual observations from 1968–2024, with no missing values in the regression variables.

The undefined first growth value is created by the transformation rather than an unreported investment observation. Other excluded years reflect differences in source coverage. Zero missing values in the final merged sample do not establish that all original data were complete.

MCAR, MAR, and MNAR cannot be reliably distinguished from the complete merged sample alone. I therefore describe the exclusions as coverage limitations rather than claim that missingness is random. The results apply to the observed study period.

Table 1. Summary Statistics for the Regression Sample

Variable	N	Mean	Standard deviation	Minimum	Maximum
Real investment growth (%)	57	3.84	5.98	−15.99	16.19
Federal funds rate (%)	57	4.98	3.77	0.08	16.38
Inflation (%)	57	4.05	2.80	−0.36	13.55
Real GDP growth (%)	57	2.80	2.02	−2.60	7.20

Annual U.S. observations from 1968–2024. The federal funds rate is an annual average; the other variables are annual growth or inflation rates.

Real investment growth varies substantially across years. Its standard deviation exceeds its mean, and the sample includes both large contractions and strong growth. The wide range of interest rates indicates that the analysis combines years with very different monetary conditions.

Descriptive Evidence

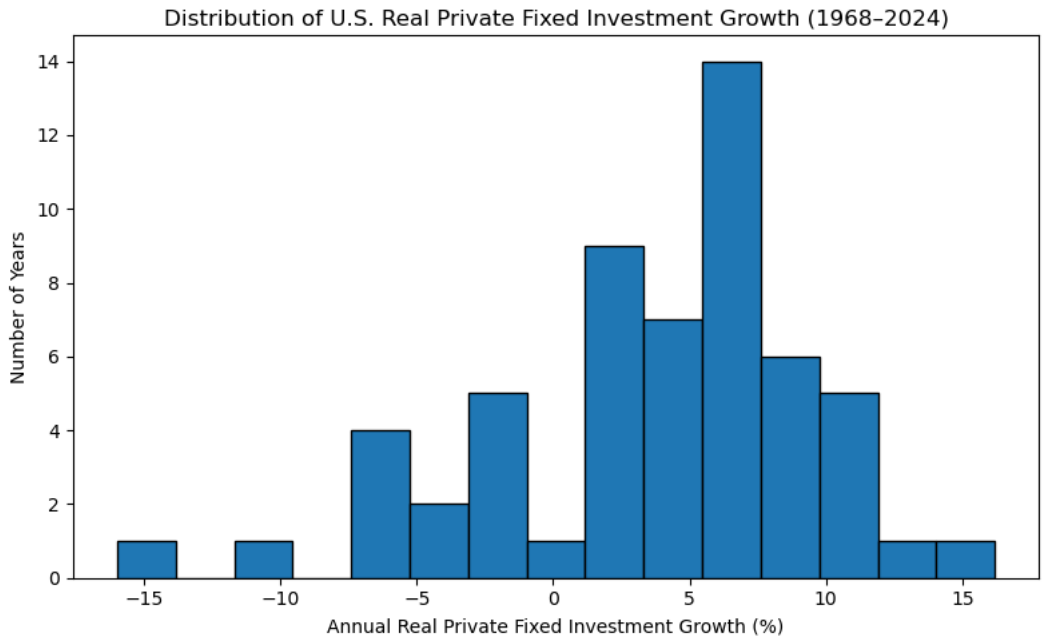


Figure 1. Real private fixed investment growth is concentrated around positive values, with several unusually low-growth years. The distribution is left-skewed, with skewness of −0.91.

I retain the growth-rate scale because it is directly interpretable and includes economically meaningful contractions. An ordinary logarithm of the growth variable would be undefined for negative observations. Outcome skewness alone does not determine whether OLS is appropriate; the residual diagnostics provide additional information about the fitted model.

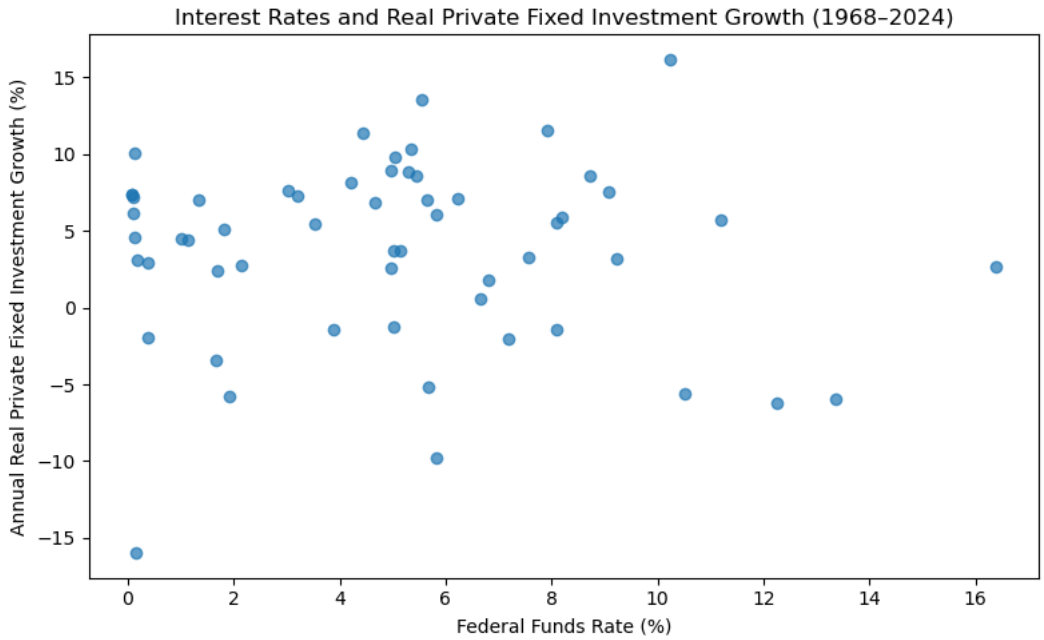


Figure 2. Investment growth varies substantially across interest-rate levels, with little apparent linear association before accounting for inflation and GDP growth.

The scatterplot does not show a strong simple relationship between higher interest rates and lower investment growth. However, it compares years that may differ in economic activity and inflation. The regression analysis examines how the estimated association changes when these conditions are included as controls. Neither the scatterplot nor the regressions establish causation.

Model and Interpretation

I estimate three OLS specifications. V1 includes only the federal funds rate, V2 adds inflation, and V3 adds real GDP growth.

The full specification is:

$$\text{Investment growth}_t = \beta_0 + \beta_1 \text{Interest rate}_t + \beta_2 \text{Inflation}_t + \beta_3 \text{GDP growth}_t + \varepsilon_t$$

The coefficient of interest is β_1. It describes the association between the federal funds rate and real investment growth conditional on the included controls.

Control Choices

Inflation is a potential confounder because it may influence monetary policy and real investment decisions. However, interest rates can also affect inflation. Contemporaneous inflation may therefore be partly downstream of interest rates, making it a potentially problematic control for estimating a total causal effect.

GDP growth accounts for broader economic activity that may influence both policy rates and investment. It can also explain variation in the outcome. Nevertheless, interest rates may affect GDP growth, and investment contributes to GDP. Conditioning on GDP growth therefore complicates causal interpretation.

The specifications describe different associations: V1 is unconditional, V2 conditions on inflation, and V3 additionally conditions on GDP growth.

Table 2. OLS Models of Annual Real Private Fixed Investment Growth

Variable	V1	V2	V3
Intercept	4.222 (1.331)	4.972 (1.439)	−2.477 (0.922)
Federal funds rate	−0.076 (0.214)	0.214 (0.306)	−0.196 (0.156)
Inflation	—	−0.543 (0.412)	0.006 (0.209)
Real GDP Growth	—	—	2.596 (0.202)
Observations	57	57	57
R^2	0.002	0.034	0.765

Conventional OLS standard errors appear in parentheses. The dependent variable is annual real private fixed investment growth (%). HC3 results for V3 are discussed below.

The interest-rate coefficient changes from −0.076 in V1 to +0.214 in V2 and −0.196 in V3. Its conventional p-values are 0.723, 0.486, and 0.2137, respectively. It is not statistically significant at the 5% level in any specification.

R^2 rises substantially when GDP growth is added. This indicates greater explanatory power within the sample, but it does not establish that V3 identifies a causal relationship.

Coefficient Interpretations

In V3, a one-percentage-point increase in the annual average federal funds rate is associated with a 0.196-percentage-point decrease in annual real private fixed investment growth, holding inflation and real GDP growth constant.

The conventional standard error is 0.1556, with $p = 0.2137$. The HC3 robust standard error is 0.2137, with $p = 0.3635$. The estimate is negative, but its precision is insufficient to establish a negative conditional association at the 5% level. This does not demonstrate that the true relationship is zero.

A one-percentage-point increase in real GDP growth is associated with a 2.596-percentage-point increase in real private fixed investment growth, holding the federal funds rate and inflation constant. Its conventional standard error is 0.2023, and its HC3 standard error is 0.3518. The estimate remains statistically significant with HC3 errors ($p < 0.001$). This association should be interpreted cautiously because investment contributes to GDP.

Diagnostics

Residual Patterns

I examine V3 residuals against fitted values and against the federal funds rate. The plots do not reveal a clear curved pattern, although some observations have large residuals. The absence of an obvious curve does not establish that the specification is correct or that all OLS assumptions hold.

Heteroskedasticity and Robust Inference

The Breusch–Pagan test produces an LM statistic of 10.708 and a p-value of 0.0134. I reject the null hypothesis of constant variance at the 5% level. This supports using heteroskedasticity-robust standard errors.

HC3 standard errors are larger than conventional standard errors for all V3 coefficients. For the interest-rate coefficient, the p-value increases from 0.2137 to 0.3635. The conclusion remains that the evidence is insufficient to establish a negative conditional association.

GDP growth remains statistically significant, while inflation is not significant using HC3 errors ($p = 0.9825$). Robust errors change estimated precision rather than the OLS coefficients themselves.

Multicollinearity

The VIF values are 2.17 for the federal funds rate, 2.16 for inflation, and 1.05 for GDP growth. These are below the common screening threshold of 5, suggesting that severe multicollinearity is not a major concern.

However, low VIF values do not establish that the controls are appropriate for causal interpretation. Multicollinearity and control selection are separate issues.

Influential Observations

Using the Cook's distance screening threshold of $4/n$ gives a cutoff of 0.0702. Five years exceed it: 1975, 1982, 2009, 2020, and 2021.

The largest Cook's distances are 2009 (0.4136), 2021 (0.3389), and 2020 (0.2139). These observations may have substantial influence on the fitted model. I retain them because they represent real economic conditions; exceeding a screening threshold does not by itself justify deletion.

I have not estimated an exclusion sensitivity model. The influence analysis therefore identifies observations requiring caution without establishing how their removal would change the interest-rate coefficient.

Threats and Conclusion

A causal interpretation would require interest rates to be unrelated to unobserved determinants of investment growth after conditioning on appropriate controls. This analysis does not establish that condition. The Federal Reserve responds to economic conditions, and omitted factors such as credit availability and business confidence may influence both interest rates and investment.

Inflation and GDP growth may also respond to interest rates. Controlling for them could remove part of the relationship being studied. Investment's contribution to GDP creates an additional interpretation problem.

Annual errors may be correlated across time. HC3 errors address heteroskedasticity but do not address serial correlation. Contemporaneous annual models also do not capture delayed investment responses to earlier interest rates.

The results should therefore be interpreted as conditional associations. V3 estimates a negative interest-rate coefficient, but the estimate is not statistically significant using either conventional or HC3 robust errors. The findings provide insufficient evidence of a negative conditional association in this sample rather than demonstrating that interest rates have no effect.

With another month, I would investigate lagged interest rates, additional confounders, and serial correlation. I would also examine sensitivity to influential years while retaining the full sample as the main analysis.

Data Sources

Federal Reserve Bank of St. Louis, FRED. Real Private Fixed Investment (FPICA). https://fred.stlouisfed.org/series/FPICA

Federal Reserve Bank of St. Louis, FRED. Federal Funds Effective Rate (FEDFUNDS). https://fred.stlouisfed.org/series/FEDFUNDS

Federal Reserve Bank of St. Louis, FRED. Real GDP Growth (A191RL1A225NBEA). https://fred.stlouisfed.org/series/A191RL1A225NBEA

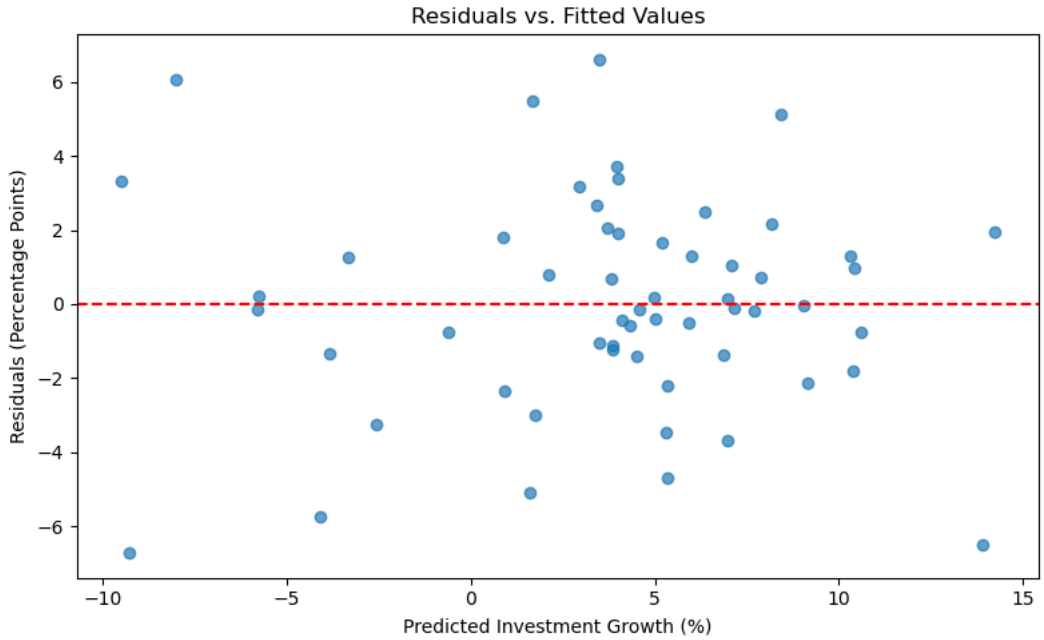
Federal Reserve Bank of St. Louis, FRED. Inflation (FPCPITOTLZGUSA). https://fred.stlouisfed.org/series/FPCPITOTLZGUSA

Appendix

Supporting Diagnostics for Version 3

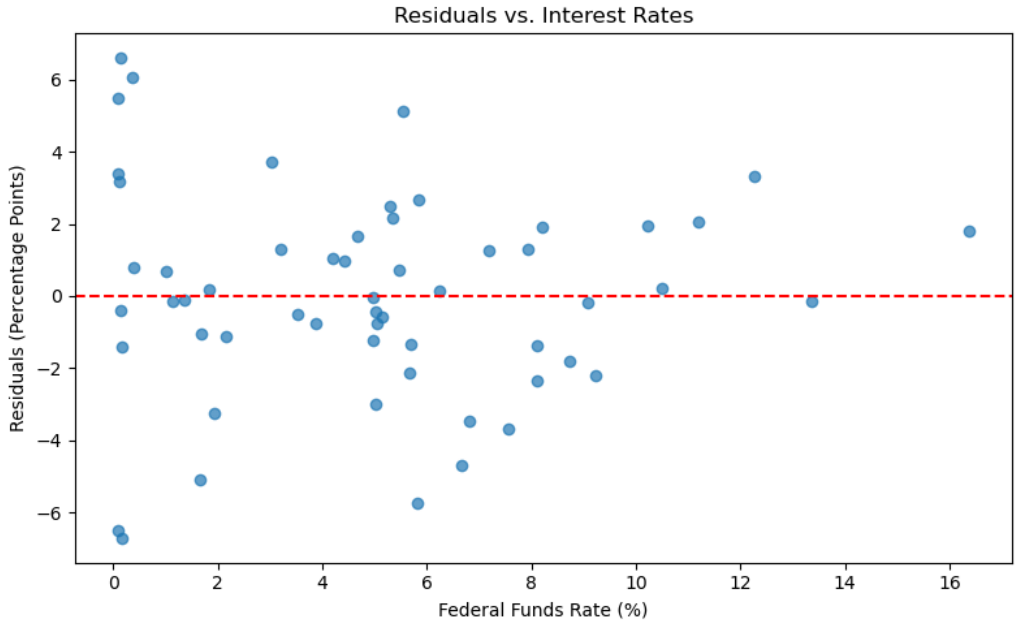
The following figures and tables provide supporting diagnostic evidence for the full OLS model. The sample contains 57 annual U.S. observations from 1968–2024.

Figure A1. Residuals Against Fitted Values



Residuals do not show a clear curved pattern, although several observations have large residuals. This visual check does not establish that all model assumptions hold.

Figure A2. Residuals Against the Federal Funds Rate



Residuals vary around zero across interest-rate levels. The Breusch–Pagan test provides a separate formal check of constant variance.

Table A1. Conventional and HC3 Inference for Version 3

Variable	Conventional SE	HC3 SE	Conventional p-value	HC3 p-value
Intercept	0.9222	1.5960	0.0096	0.1266
Federal funds rate	0.1556	0.2137	0.2137	0.3635
Inflation	0.2094	0.2848	0.9762	0.9825
Real GDP growth	0.2023	0.3518	<0.001	<0.001

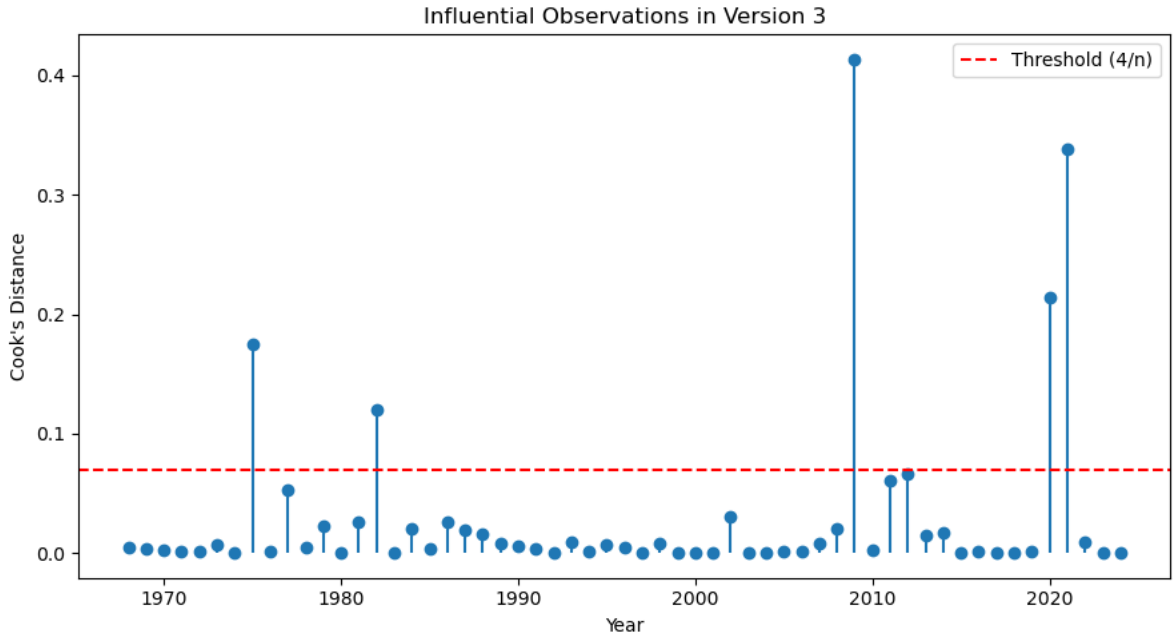
HC3 standard errors adjust inference for heteroskedasticity. They do not address serial correlation. The Breusch–Pagan LM statistic is 10.708, with p = 0.0134.

Table A2. Variance Inflation Factors

Variable	VIF
Federal funds rate	2.17
Inflation	2.16
Real GDP growth	1.05

These values suggest that severe multicollinearity is not a major concern. They do not establish that the controls are appropriate for causal interpretation.

Figure A3. Cook’s Distance by Year



The dashed line marks the screening threshold of 4/n = 0.0702. Five observations exceed this threshold.

Table A3. Observations Above the Cook's Distance Threshold

Year	Cook's Distance
2009	0.4136
2021	0.3389
2020	0.2139
1975	0.1745
1982	0.1206

All observations remain in the main model. Exceeding the screening threshold identifies potentially influential observations and does not, by itself, justify deletion. No exclusion sensitivity model was estimated.